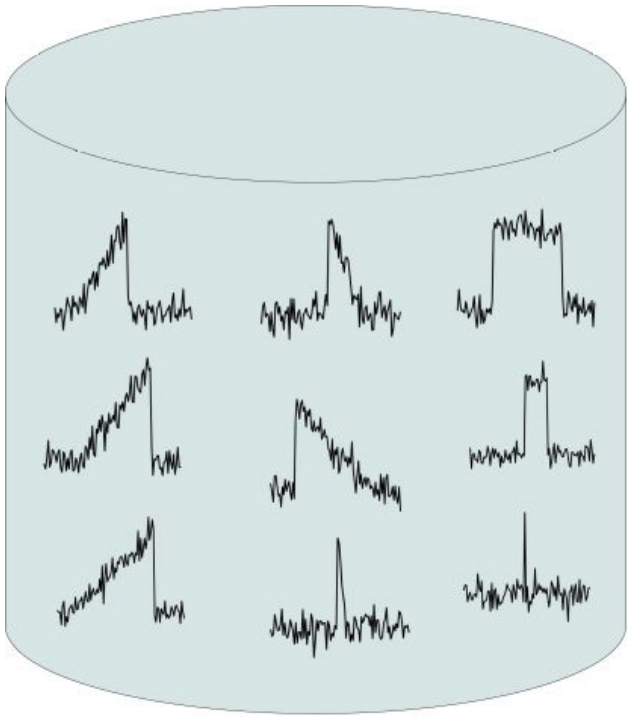
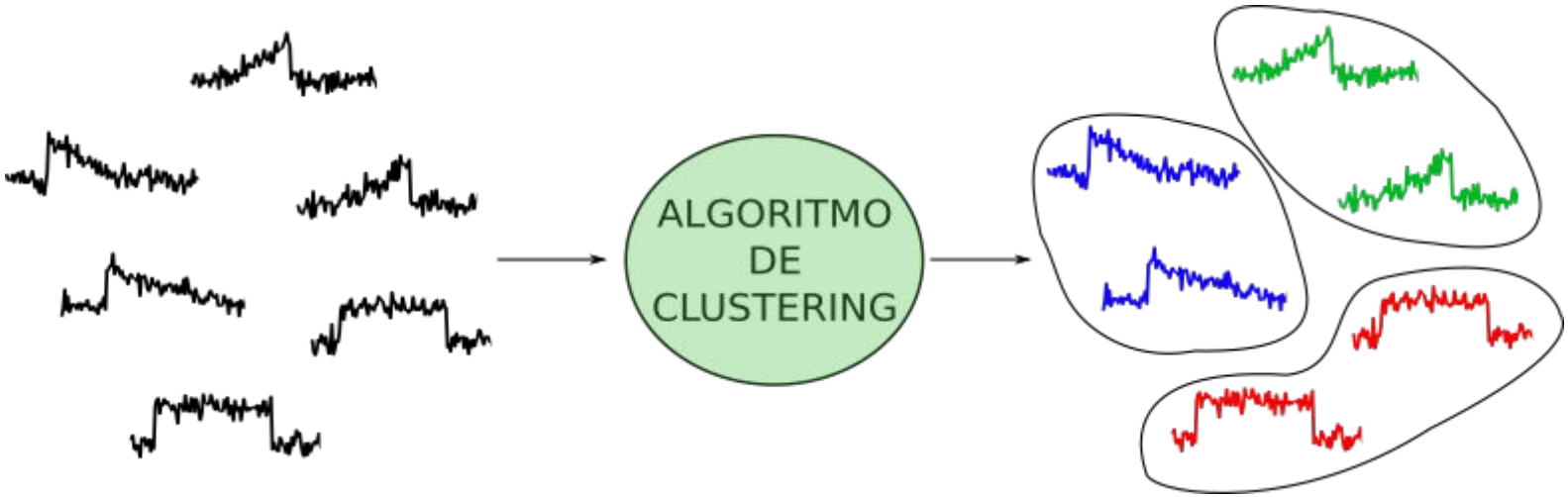
TIME SERIES CLUSTERING

Database of time series

- A set of unordered time series.
- The lengths of the time series can vary.



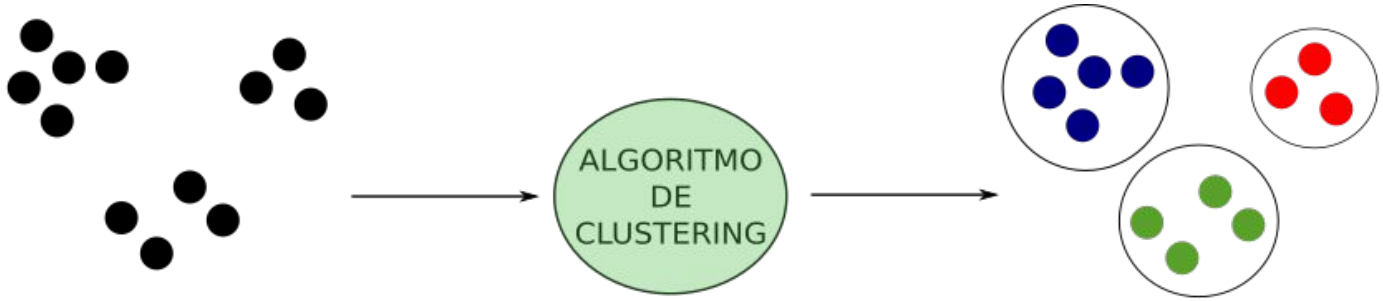
Problem definition

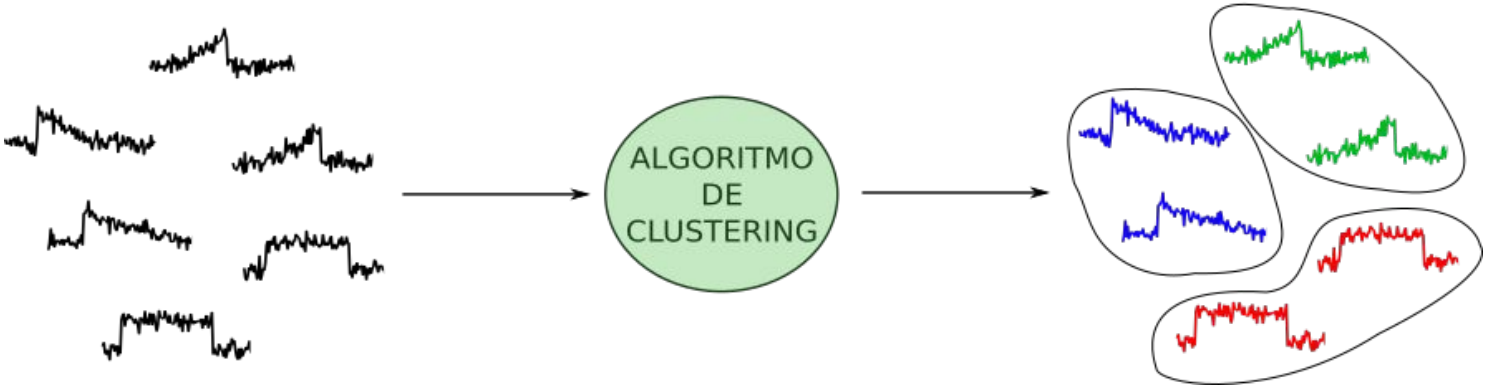


TAXI-METER
TAXI-METER
TAXI-METER

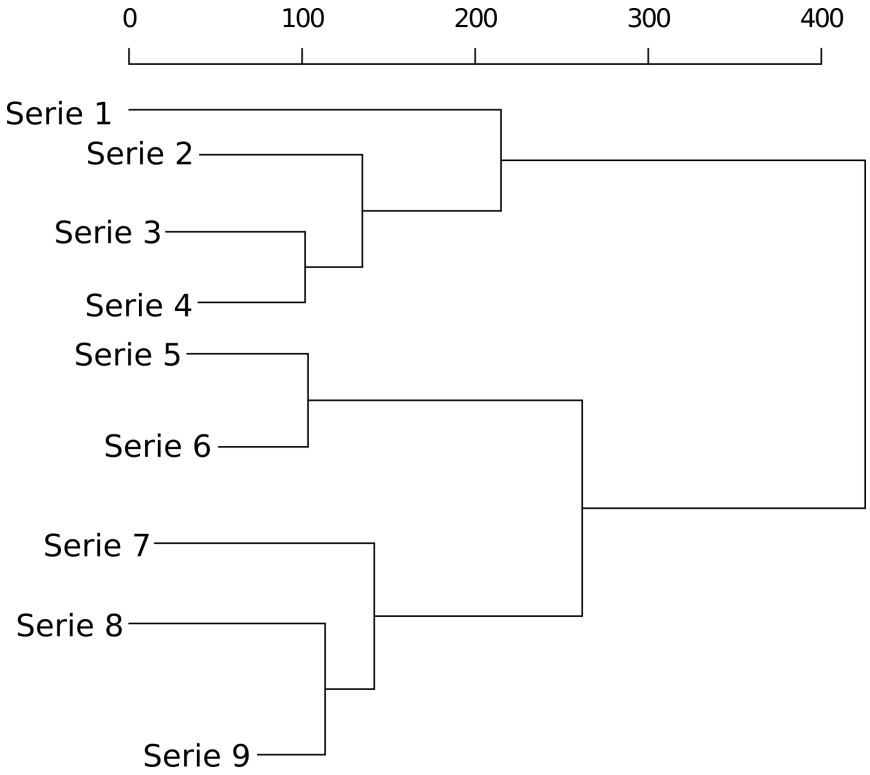
ΜΟΝΟΦΑΣΙΚΟΣ ΜΕΤΡΗΤΗΣ
kWh
HELLAS
CM143
230 V
15-60 A
50~
ΧΡ. 1997
ΔΗΟΚΤΗΣΙΑ
ΑΡΙΘΜΟΣ
ΔΕΗ
ΠΡΟΣΟΧΗ
ΡΕΥΜΑ ΚΟΜΝΕΝΟ

Difference with the classical problem





Hierarchical clustering algorithms



Partitional algorithms: k-means

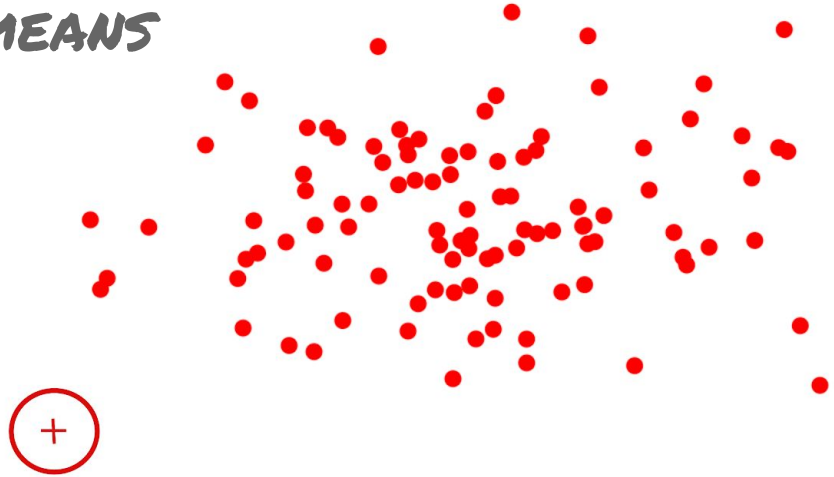
It is applied identically but now,

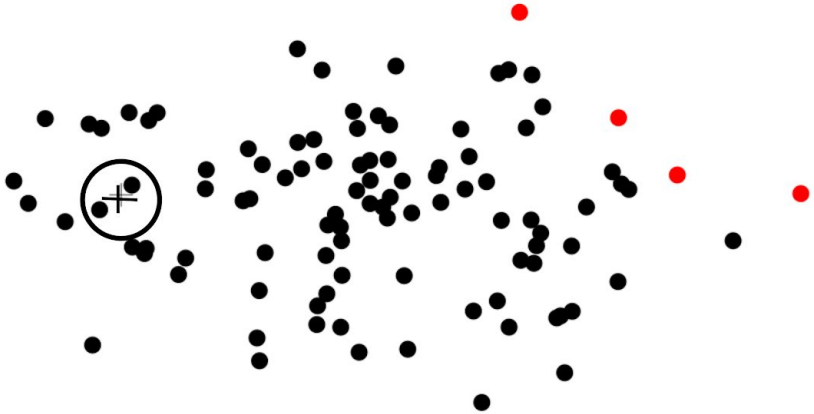
each point is a time series!

Partitional algorithms: k-means

It is applied identically but now,

each point is a time series!



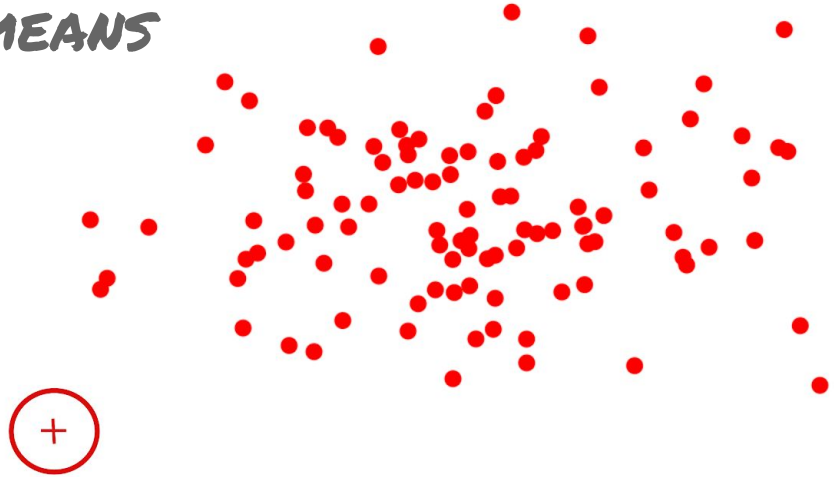


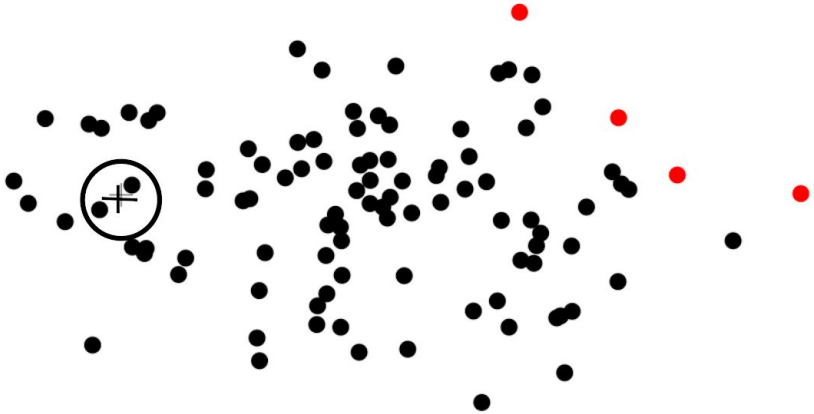
The centroids are calculated using the mean of the observations in the cluster.

Partitional algorithms: k-means

It is applied identically but now,

each point is a time series!





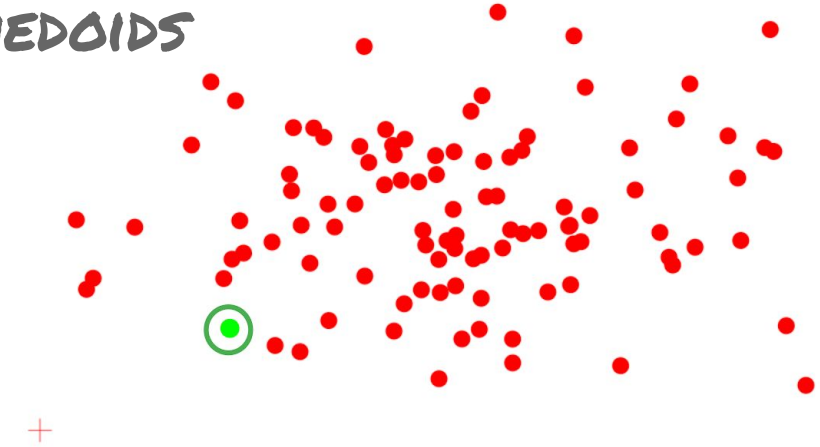
Sometimes, calculating the mean of a set of time series does not make much sense.

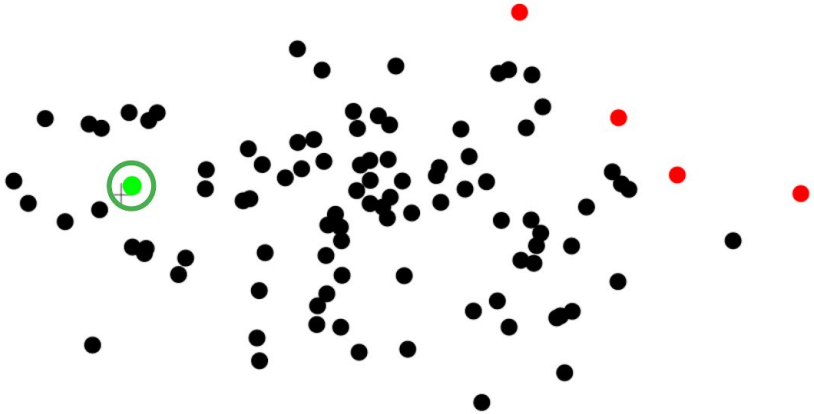
Partitional algorithms: k-medoids

Very similar to k-means

but the centroids are existing

data points.



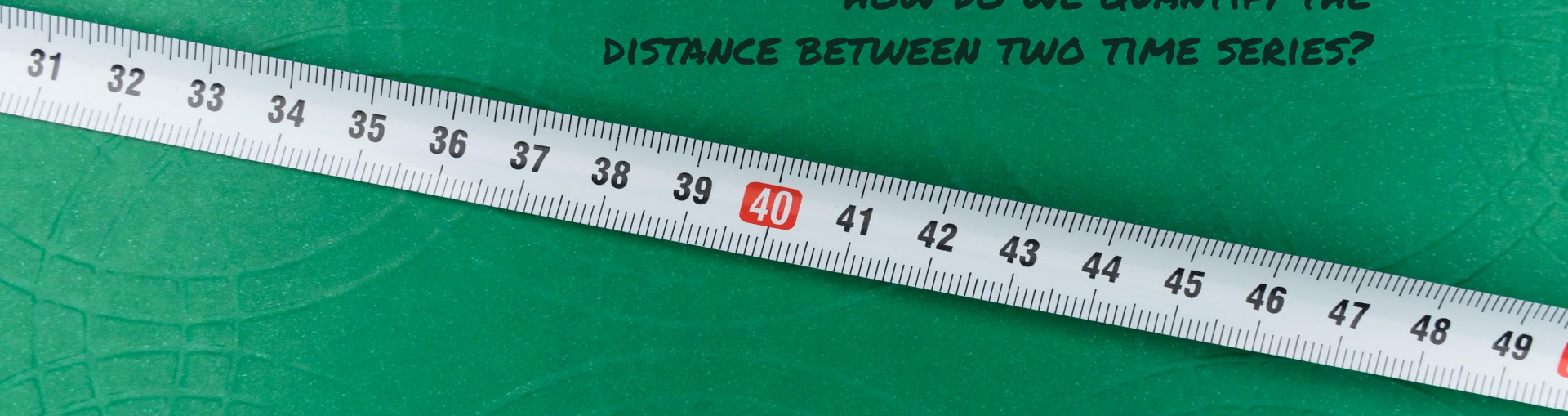


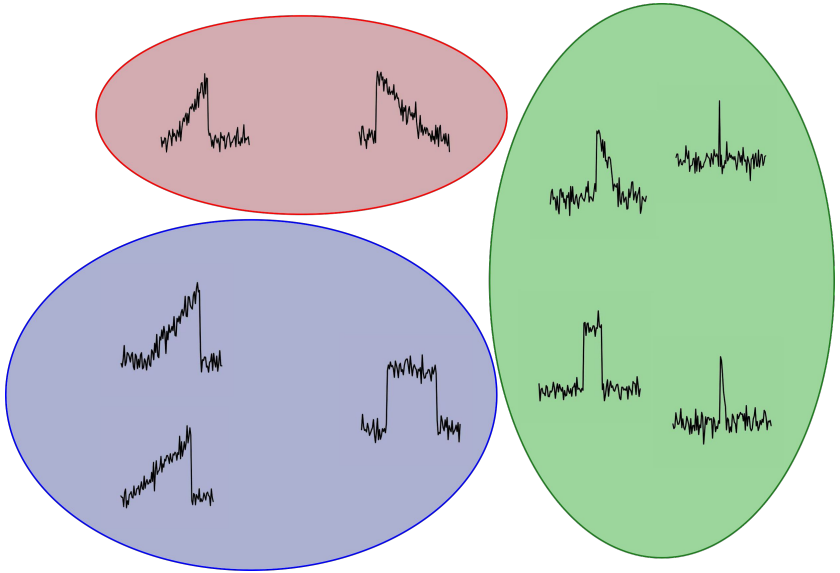
Not necessary to calculate

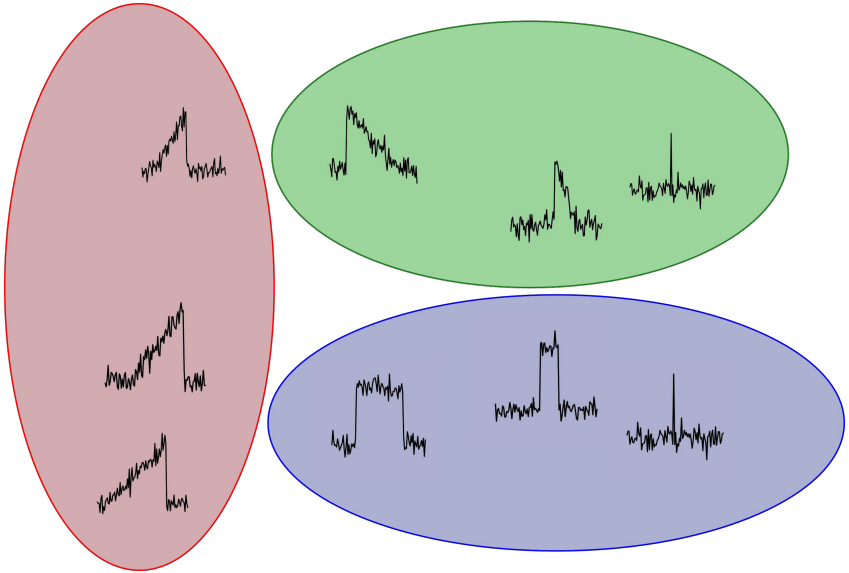
the mean!!

ALL THESE ALGORITHMS ARE BASED ON DISTANCES BUT...

HOW DO WE QUANTIFY THE DISTANCE BETWEEN TWO TIME SERIES?



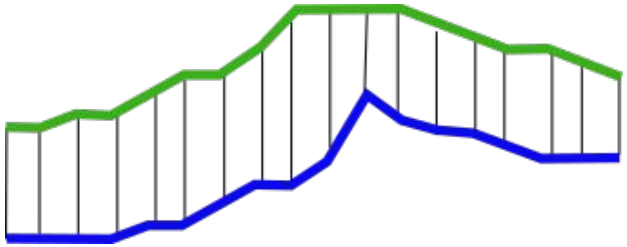




DISTANCES FOR TIME SERIES

Distances between raw time series

Fixed distances

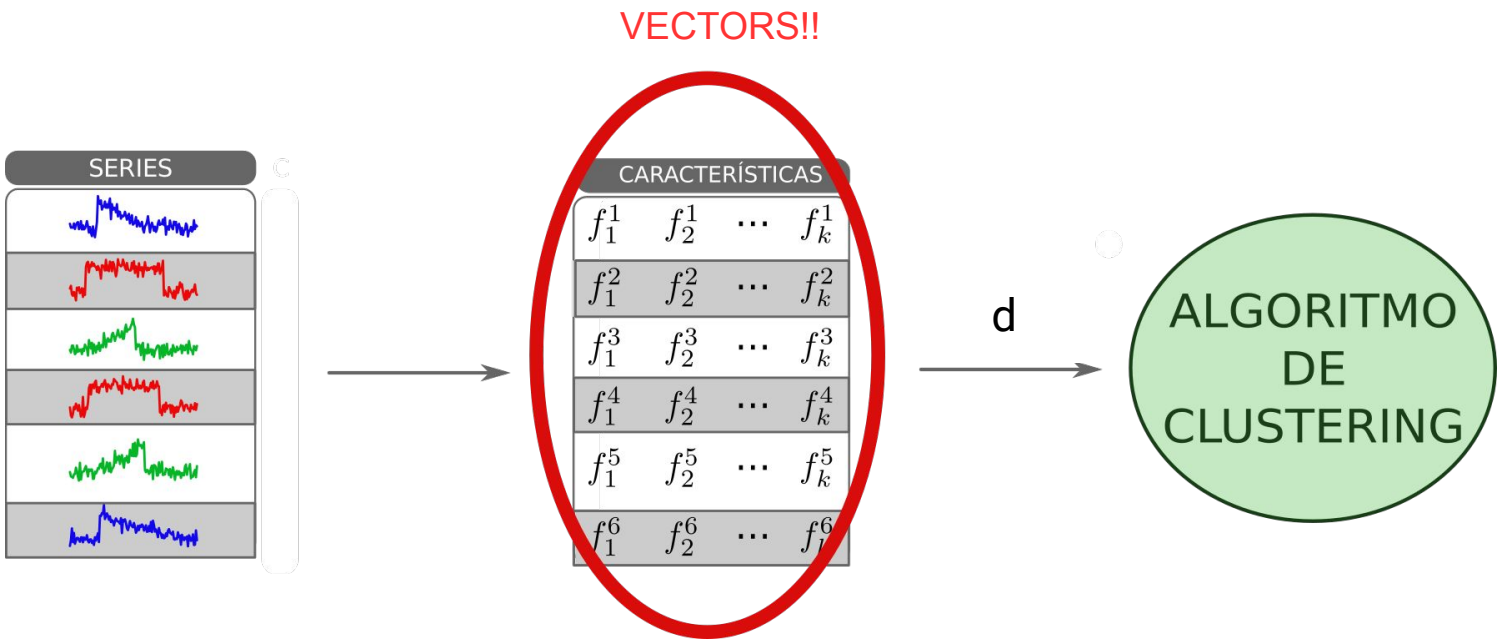


Flexible distances

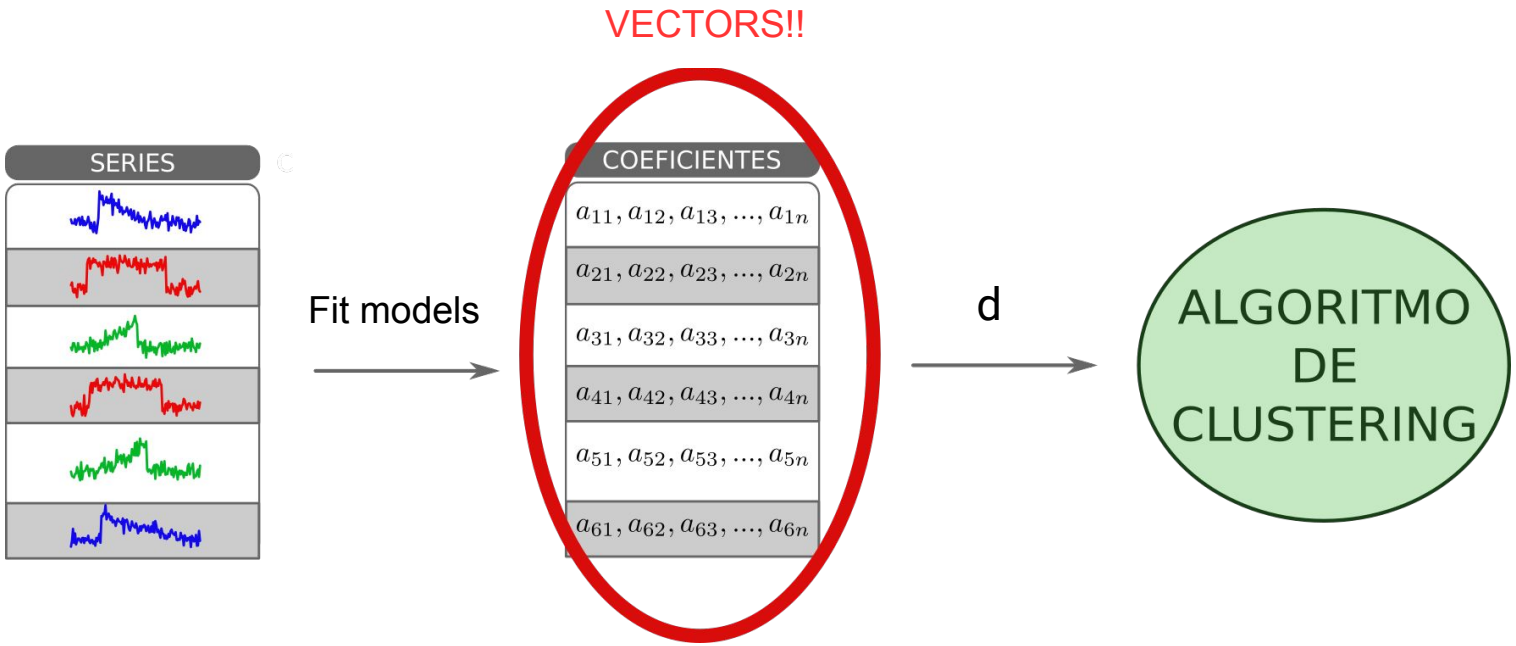
Other ways of calculating distances between time series

Represent the time series by means of **vectors** and use common distances.

Feature based clustering



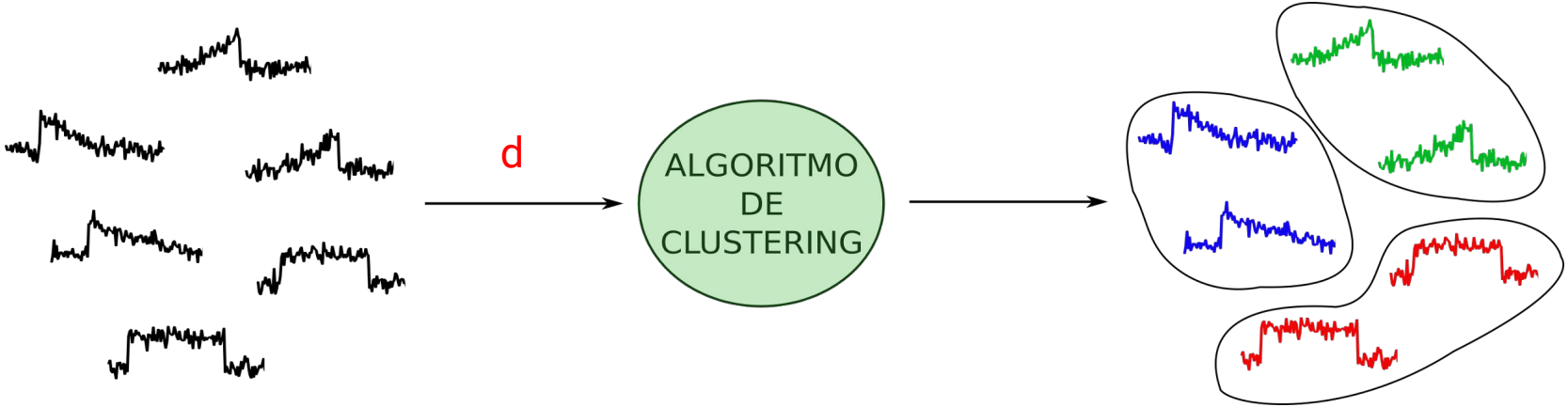
Model based clustering



Difficulties

- There is no distance measure that is the best in all the cases.
- In each case we must select a distance measure that represents the dissimilarity that we want to capture.
- How to select?
  - Domain knowledge.
  - Run several tests.

Now... back to clustering



Evaluation

- Evaluation is frequently quite subjective, and consists of visual inspection.

- There are some evaluation indices:
  - Davies–Bouldin index
  - Silhouette
  - Dunn index

✗ … but they are not always valid to compare distance measures.

Software

- Distances for time series:

  TSclust
  TSdist
  dtw

- Clustering algorithms:

  pam
  hclust

Software

- Distances for time series:

  sklearn.metrics

- Clustering algorithms:

  sklearn.cluster

  aeon (specific for time series)

Bibliography

- Philippe Esling and Carlos Agon. 2012. Time-series data mining. ACM Comput. Surv. 45, 1, Article 12 (December 2012), 34 pages.

- T. W. Liao, Clustering of time series data: a survey, Pattern Recognition, vol. 38, no. 11, pp. 1857–1874, Nov. 2005.